

The Euclidean Steinitz–Bergström theorem

OpenAI

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Abstract

Every prescribed-order finite sequence of vectors in the Euclidean unit ball of $\mathbb{R}^d$ has one signing for which every signed prefix has norm at most $C\sqrt{d}$, with C absolute and independent of the sequence length. Consequently, every indexed zero-sum family of unit-ball vectors admits an ordering with the same bound for its unsigned partial sums. This determines the Euclidean Steinitz constant up to absolute factors, $S_2(d) = \Theta(\sqrt{d})$, and resolves the Euclidean Steinitz–Bergström conjecture.

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1 Introduction

Let $v_1, \dots, v_N$ be vectors in the Euclidean unit ball of $\mathbb{R}^d$. There are two natural ways to keep their partial sums small: choose signs in a prescribed order, or, when the family sums to zero, choose an order without changing any signs. The difficulty in the first problem is to control every prefix by one choice of signs, with a bound independent of N . We prove such a bound on the square-root scale in the dimension.

Theorem 1.1 (Prescribed-order signed prefixes). *There is an absolute constant C such that, for all integers $d, N \geq 1$ and every sequence $v_1, \dots, v_N \in \mathbb{R}^d$ with $\|v_i\|_2 \leq 1$, there are signs $\varepsilon_1, \dots, \varepsilon_N \in \{-1, 1\}$ for which*

$$\max_{0 \leq k \leq N} \left\| \sum_{i=1}^k \varepsilon_i v_i \right\|_2 \leq C\sqrt{d}.$$

The same signs control all prefixes, and C is independent of d , N and the sequence.

The zero-sum ordering problem is quantified by the Euclidean Steinitz constant. For an indexed family with $\sum_{i=1}^N v_i = 0$, define

$$\beta(v_1, \dots, v_N) = \min_{\pi \in \mathfrak{S}_N} \max_{0 \leq k \leq N} \left\| \sum_{i=1}^k v_{\pi(i)} \right\|_2,$$

where $\mathfrak{S}_N$ is the set of permutations of $\{1, \dots, N\}$ and an empty sum is zero. Let $S_2(d)$ be the supremum of β over all finite indexed zero-sum families of Euclidean unit-ball vectors in $\mathbb{R}^d$. The Euclidean Steinitz–Bergström conjecture asks whether $S_2(d) = O(\sqrt{d})$. Theorem 1.1 gives its affirmative resolution.

Theorem 1.2 (Euclidean Steinitz–Bergström bound). *There is an absolute constant C such that, for all integers $d, N \geq 1$ and every indexed family $v_1, \dots, v_N \in \mathbb{R}^d$ satisfying $\|v_i\|_2 \leq 1$ and $\sum_{i=1}^N v_i = 0$, some permutation $\pi \in \mathfrak{S}_N$ satisfies*

$$\max_{0 \leq k \leq N} \left\| \sum_{i=1}^k v_{\pi(i)} \right\|_2 \leq C\sqrt{d}.$$

The constant in Theorem 1.1 also suffices here.

Both statements allow repetitions and zero vectors. The empty-sequence and dimension-zero extensions have every partial sum equal to zero. The construction is existential: its choices may depend on the complete input sequence. Its successive choice of signs does not supply an online rule or an efficient algorithm.

The order of growth in Theorem 1.2 is necessary. In the d -dimensional hyperplane orthogonal to the all-ones vector $\mathbf{1} \in \mathbb{R}^{d+1}$, let e_i denote the standard coordinate vectors and take

$$u_i = \sqrt{\frac{d+1}{d}} \left(e_i - \frac{\mathbf{1}}{d+1} \right), \quad 1 \leq i \leq d+1.$$

These unit vectors sum to zero and satisfy $\langle u_i, u_j \rangle = -1/d$ for $i \neq j$. Hence every sum of k distinct members has squared norm $k(d+1-k)/d$. With $k = \lfloor (d+1)/2 \rfloor$, this is at least $d/4$, regardless of the ordering. Thus

$$\frac{1}{2}\sqrt{d} \leq S_2(d) \leq C\sqrt{d} \quad (d \geq 1).$$

1.1 From signs to an ordering

We give the short transference argument at once. Chobanyan’s transference principle [13] relates rearrangements and signed sums. We use its finite positive-forward, negative-reverse form [12, Theorem 2.1 and Remark 1].

Proof of Theorem 1.2 from Theorem 1.1. Fix an indexed zero-sum family and choose an order attaining its minimum β ; this exists because there are only finitely many permutations. Relabel the vectors in that order. Put

$$A_j = \sum_{i=1}^j v_i, \quad B_j = \sum_{i=1}^j \varepsilon_i v_i, \quad 0 \leq j \leq N,$$

where the signs come from Theorem 1.1. Hence $\|A_j\|_2 \leq \beta$ and $\|B_j\|_2 \leq C\sqrt{d}$. The positive-sign and negative-sign selections from an original prefix sum respectively to

$$P_j = \frac{A_j + B_j}{2}, \quad M_j = \frac{A_j - B_j}{2}.$$

Place all positive-sign indices in their original order, followed by all negative-sign indices in reverse original order. A partial sum in the first block is some P_j . In the second block, the unused indices are the negative-sign indices of an original prefix. Since the total sum is zero, the current partial sum is therefore some $-M_j$. This also covers either block being empty. Every new partial sum has norm at most $(\beta + C\sqrt{d})/2$. By minimality,

$$\beta \leq \frac{\beta + C\sqrt{d}}{2}, \quad \text{and hence} \quad \beta \leq C\sqrt{d}.$$

This permutes indices and so preserves all multiplicities. $\square$

The signed-prefix bound also yields finite-dimensional ℓ_p estimates and a colorful row-permutation theorem. We develop these distinct consequences in Section 6.

1.2 Historical context

The finite rearrangement theorem goes back to Steinitz’s work on conditionally convergent vector series; Ambrus and Heck give a historical account in [2, Section 2]. The dimension bound for zero-sum rearrangement holds for arbitrary norms: Grinberg and Sevast’yanov [16, Theorem 1] prove that the partial sums can be kept within d times the unit ball. The smaller square-root scale in Euclidean space has a long history; Behrend discusses this expected growth in [8, p. 108]. The modern formulation and its attribution to Bergström are recorded by Ambrus and Heck [2, Conjecture 5]. Their Theorem 7 reduces the asymptotic question to a relaxed rearrangement problem for vectors with norms in $[1 - \varepsilon, 1]$, with $0 < \varepsilon \leq 1$ fixed independently of the dimension. In that problem each prefix is compared with its proportional share of the total sum, rather than with zero. For prescribed-order Euclidean signing, the square-root-dimension question is stated explicitly by Bansal, Jiang, Meka, Singla and Sinha [6, Section 6, Conjecture 6.3], following Banaszczyk [4]. Their discussion records Banaszczyk’s bound $O(\sqrt{d} + \sqrt{\log N})$; the length term becomes significant when N is large compared with the dimension. In a 2026 preprint, Dutta, Jha and Jiang state an efficient prescribed-order signing bound

$$O(\sqrt{d} + d^{1/4} \log^{7/4} N),$$

and the corresponding zero-sum ordering bound [15, Theorem 1.1 and Corollary 1.2]. Theorem 1.1 gives an absolute multiple of $\sqrt{d}$ for every finite length.

Terminal vector balancing has a different quantifier. For columns in the Euclidean unit ball, Li [18, Theorem 1.1] bounds the final signed sum in infinity norm by an absolute constant and gives a deterministic polynomial-bit algorithm on rational input. That terminal statement implies a Euclidean bound of order $\sqrt{d}$ for one sum. Applying it separately to each prefix permits the signs to change with the prefix; Theorem 1.1 instead supplies one common signing. The present all-prefix estimate is Euclidean and does not assert a constant infinity-norm all-prefix bound.

1.3 Proof strategy

Gaussian convex bodies are a central setting for vector balancing, notably in the theorem of Banaszczyk [3]. Our proof takes place in a space of coefficients: one coordinate is reserved for each incoming vector, together with d initial state coordinates. A sequence of linear maps sends these coefficients to states in $\mathbb{R}^d$. At each step an invertible contraction acts on the old state and the next coefficient injects a multiple of the incoming vector. The body of coefficients for which every state is bounded will provide the simultaneous control of all prefixes.

Two analytic ingredients make this construction possible. First, if a bounded symmetric convex body has small Dirichlet energy for every positive diagonal weight, it supports a single square root of a probability density with small derivatives in every coordinate. A lifted Steiner symmetrization preserves these bounds and permits a coordinate step of the form $q + \varepsilon$, where $|q| \leq 1/4$ and $\varepsilon \in \{-1, 1\}$. This adapts the directional-density and symmetrization method of Guo, Fang and Lu [17, Sections 3–4].¹ Bandeira [5] gives a related formulation for terminal balancing in terms of Dirichlet eigenvalues. Akbas and Sra [1, Lemmas 2.2–2.3] develop the common quadratic-energy witness and lifted L^2 -symmetrization argument. Here the hypotheses use diagonal weights, and the coordinate shifts may depend on earlier choices. We prove the required energy and shifted-step statements in full.

Second, the filtered body has a diagonal Dirichlet-energy bound independent of the number of filters. To prove this, we follow a stationary Ornstein–Uhlenbeck process in coefficient space and localize its filtered states to dyadic spectral scales. A matrix estimate controls changes at one scale by an additive trace budget, including gains inserted between noncommuting contractions. At a fixed scale, this estimate gives a uniform positive probability for all constraints in a group of indices with bounded accumulated energy, over a matched time interval. Gaussian correlation combines these groups, scales and time intervals; the exponential survival rate yields the energy bound.

The two ingredients meet through a scalar predictor. The allowed shift in the incoming coefficient exactly compensates for the contraction of the old state. Consequently the final states are the signed partial sums themselves. We give this complete construction in Section 2, after stating the two analytic propositions. Section 3 proves the shifted-sign proposition. Section 4 establishes the matrix variation estimate, and Section 5 uses it to prove the filtered-body energy bound. Section 6 develops the norm-comparison and colorful row-permutation consequences.

Throughout, all unspecified constants are absolute. Vectors and matrices are real; $\|\cdot\|_{\text{op}}$ and $\|\cdot\|_{\text{HS}}$ denote operator and Hilbert–Schmidt norms. For symmetric matrices, $A \leq B$ means that $B - A$ is positive semidefinite. Functions of positive definite matrices are defined by the spectral calculus.

¹Guo, Fang and Lu credit the Odin Automatic AI Research Agent for their proof.

2 Constructing the signed walk

We will encode the signed walk by one point in a convex body in a larger coefficient space. The body controls a contracted version of every prefix. Its extra coordinates will be chosen so that their contributions exactly restore the parts removed by the contractions. This section states the two analytic inputs and proves Theorem 1.1 from them.

2.1 The two analytic inputs

Let $D \subset \mathbb{R}^m$ be a nonempty bounded open convex set, and let Q be a positive definite diagonal $m \times m$ matrix. Define its diagonal Dirichlet energy by

$$\lambda_Q(D) = \inf_{0 \neq g \in H_0^1(D)} \frac{\int_D \langle Q \nabla g, \nabla g \rangle dx}{\int_D g^2 dx}. \quad (1)$$

Here $H_0^1(D)$ is the closure of $C_c^\infty(D)$ in the Sobolev norm, and derivatives are weak derivatives. This normalization corresponds to Brownian motion with covariance rate $2Q$. Central symmetry will always mean symmetry about the origin. Fix

$$\kappa = \frac{1}{100}, \quad \delta = \frac{1}{4}. \quad (2)$$

Our first input turns small coordinate energies into signs while allowing a bounded shift at each step. The shifts may depend on previously assigned coordinates.

Proposition 2.1 (Signs with adaptive shifts). *Let $K \subset \mathbb{R}^m$ be nonempty, bounded, open, convex and centrally symmetric. Suppose that*

$$\lambda_Q(K) \leq \kappa^2 \operatorname{tr} Q \quad \text{for every positive definite diagonal } Q. \quad (3)$$

Let $j_1, \dots, j_s$ be distinct elements of $\{1, \dots, m\}$, and let $q_i : \mathbb{R}^{i-1} \rightarrow [-\delta, \delta]$ be arbitrary functions, for $1 \leq i \leq s$. There exist signs $\varepsilon_1, \dots, \varepsilon_s$ such that the recursion

$$x^{(0)} = 0, \quad x^{(i)} = x^{(i-1)} + (q_i(x_{j_1}^{(i-1)}, \dots, x_{j_{i-1}}^{(i-1)}) + \varepsilon_i) e_{j_i}$$

has $x^{(s)} \in K$. For $i = 1$, the function q_1 has a one-point domain.

This is an existence statement; the choice of signs can depend on the whole body K and all the prescribed functions. Its proof uses backward transformations of K , followed by forward choices of the signs.

The second input supplies a body satisfying a uniform energy bound. Its linear maps retain a d -dimensional state and read one new coordinate at each step.

Proposition 2.2 (Energy of a filtered path body). *There are positive absolute constants A_0, B_0, H_0 with the following property. Let $d \geq 1$ and $n \geq 0$ be integers and put $m = d + n$. For $1 \leq t \leq n$, let C_t be an invertible real $d \times d$ matrix and let $b_t \in \mathbb{R}^d$ satisfy*

$$C_t C_t^T + b_t b_t^T = I_d.$$

Define linear maps $R_t : \mathbb{R}^m \rightarrow \mathbb{R}^d$ by

$$R_0 x = (x_1, \dots, x_d), \quad R_t x = C_t R_{t-1} x + b_t x_{d+t} \quad (1 \leq t \leq n). \quad (4)$$

Then the bounded open convex centrally symmetric set

$$\mathcal{D} = \left\{ x \in \mathbb{R}^m : \begin{array}{ll} |x_j| < B_0 & (1 \leq j \leq m), \\ \|R_t x\|_2 < A_0 \sqrt{d} & (0 \leq t \leq n) \end{array} \right\} \quad (5)$$

satisfies

$$\lambda_Q(\mathcal{D}) \leq H_0 \operatorname{tr} Q \quad \text{for every positive definite diagonal } Q.$$

The constants are independent of the number of maps and of Q . This uniformity is essential: the application will use n equal to the number of vectors. To compensate for the contractions, we will also impose linear constraints on the coefficients. The following estimate bounds their additional energy cost; we prove it after completing the construction.

Lemma 2.3 (Adding slabs). *Let $D \subset \mathbb{R}^m$ be bounded, nonempty, open, convex and centrally symmetric. Let $D_* > 0$, $\delta > 0$, and let $m_1, \dots, m_n$ be real rows of length m . Set*

$$K = (2D_* D) \cap \bigcap_{i=1}^n \{x : |m_i x| < \delta\}.$$

For every positive definite diagonal Q ,

$$\lambda_Q(K) \leq D_*^{-2} \lambda_Q(D) + \pi^2 \delta^{-2} \sum_{i=1}^n m_i Q m_i^T. \quad (6)$$

2.2 The filter, the shifts, and their cancellation

We now prove the signed-prefix theorem from Propositions 2.1 and 2.2. Their proofs in the remaining sections are independent of this application.

Proof of Theorem 1.1. Write the prescribed sequence as $v_1, \dots, v_n \in \mathbb{R}^d$, with $\|v_i\|_2 \leq 1$. The hypotheses give $d, n \geq 1$.

The symmetric contractions. For a positive absolute $\sigma < 1$, to be chosen below, set

$$b_i = \sigma v_i, \quad \eta_i = \frac{1}{1 + \sqrt{1 - \|b_i\|_2^2}}, \quad C_i = I - \eta_i b_i b_i^T. \quad (7)$$

On the span of b_i , the matrix C_i has eigenvalue $\sqrt{1 - \|b_i\|_2^2}$, and on its orthogonal complement it is the identity. This includes $b_i = 0$, for which $\eta_i = 1/2$ and $C_i = I$. Consequently

$$C_i = (I - b_i b_i^T)^{1/2} = C_i^T \succ 0, \quad C_i^2 + b_i b_i^T = I, \quad \frac{1}{2} \leq \eta_i \leq 1.$$

Use these matrices in (4), and let $\mathcal{D}$ be the body in Proposition 2.2.

For $x \in \mathbb{R}^{d+n}$, let x^0 be obtained by setting its first d coordinates to zero. Define row vectors m_i by

$$m_i x = \eta_i b_i^T R_{i-1} x^0. \quad (8)$$

Each row m_i is supported only on the noise coordinates $d+1, \dots, d+i-1$. In particular it is unaffected by any coordinate assigned at step i or later.

The cost of the predictor constraints. Let M be the matrix with rows m_i . We claim that every column of M has squared Euclidean norm at most σ^2 . Its initial d columns are zero. For a noise coordinate $d + j$, follow its contribution to the state:

$$r_j = b_j, \quad r_i = C_i r_{i-1} \quad (j < i \leq n).$$

The coefficient of that coordinate in m_i is zero for $i \leq j$, and is $\eta_i b_i^T r_{i-1}$ for $i > j$. Symmetry of C_i gives the exact loss identity

$$\|r_{i-1}\|_2^2 - \|r_i\|_2^2 = (b_i^T r_{i-1})^2.$$

Thus the losses telescope:

$$\begin{aligned} \|M e_{d+j}\|_2^2 &= \sum_{i=j+1}^n \eta_i^2 (b_i^T r_{i-1})^2 \\ &\leq \sum_{i=j+1}^n (\|r_{i-1}\|_2^2 - \|r_i\|_2^2) \leq \|b_j\|_2^2 \leq \sigma^2. \end{aligned} \tag{9}$$

In particular, for every positive definite diagonal Q ,

$$\sum_{i=1}^n m_i Q m_i^T = \sum_{j=1}^{d+n} Q_{jj} \|M e_j\|_2^2 \leq \sigma^2 \operatorname{tr} Q. \tag{10}$$

Choose an absolute dilation $D_* > 0$ and define

$$K = (2D_* \mathcal{D}) \cap \{x : |m_i x| < \delta \text{ for } 1 \leq i \leq n\}. \tag{11}$$

This is a bounded open convex centrally symmetric set containing a neighborhood of zero. Lemma 2.3, Proposition 2.2, and (10) give

$$\lambda_Q(K) \leq (H_0 D_*^{-2} + \pi^2 \delta^{-2} \sigma^2) \operatorname{tr} Q. \tag{12}$$

We can therefore fix, once and for all,

$$D_* \geq \max\{1, \sqrt{2H_0/\kappa}\}, \quad \sigma = \min\{1/2, \kappa\delta/(\sqrt{2\pi})\}. \tag{13}$$

The same body K then satisfies (3) for every positive definite diagonal Q . These choices are independent of d, n and the vectors.

Choosing the coefficients. For $u \in \mathbb{R}$ write $\operatorname{clip}_\delta(u) = \max\{-\delta, \min\{u, \delta\}\}$. Apply Proposition 2.1 to the distinct axes $e_{d+1}, \dots, e_{d+n}$. At step i prescribe the shift

$$q_i = \operatorname{clip}_\delta(m_i x^{(i-1)}).$$

By the support of m_i , this is a function of the previously assigned coordinates, as required. The proposition supplies signs and a final point $x = x^{(n)} \in K$. Its first d coordinates remain zero. Also, coordinates assigned before step i never change afterward, so

$$m_i x^{(i-1)} = m_i x.$$

The final membership $x \in K$ implies $|m_i x| < \delta$. Hence each shift was already within the permitted interval: clipping was inactive, and

$$x_{d+i} = m_i x + \varepsilon_i = \eta_i b_i^T R_{i-1} x + \varepsilon_i. \tag{14}$$

Here $x^0 = x$ because the initial block is zero. Clipping is used only to make the application of Proposition 2.1 unconditional; causality and the final constraints subsequently remove it.

Figure 1 illustrates the identity that completes the construction.

Cancellation. For this single final coefficient vector, $R_0x = 0$ and

$$\begin{aligned} R_t x &= C_t R_{t-1} x + b_t x_{d+t} \\ &= (I - \eta_t b_t b_t^T) R_{t-1} x + b_t (\eta_t b_t^T R_{t-1} x + \varepsilon_t) \\ &= R_{t-1} x + \varepsilon_t b_t. \end{aligned}$$

Thus, simultaneously for every $0 \leq t \leq n$,

$$R_t x = \sum_{i=1}^t \varepsilon_i b_i. \quad (15)$$

Since $x \in 2D_*\mathcal{D}$, all these states have norm less than $2D_*A_0\sqrt{d}$. Dividing (15) by σ proves Theorem 1.1 with the absolute constant $C = 2D_*A_0/\sigma$. $\square$

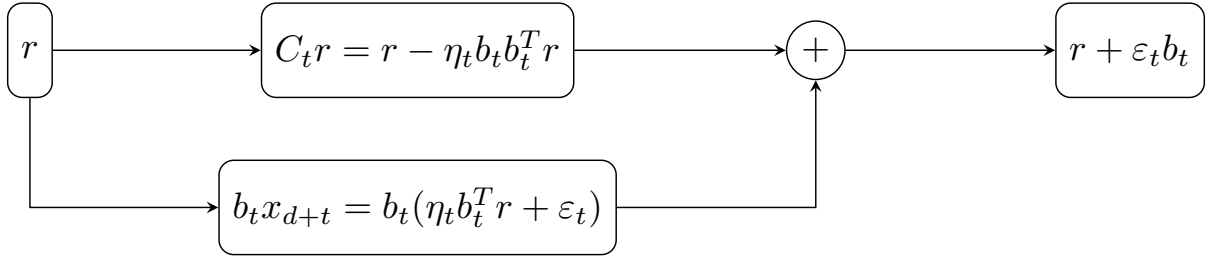


Figure 1: For the final coefficient vector, put $r = R_{t-1}x$. The predictor term in the incoming coordinate restores exactly the component removed by C_t , leaving the signed increment $\varepsilon_t b_t$.

2.3 Brownian proofs of the slab bound

We now prove Lemma 2.3, which bounded the cost of the predictor constraints. Brownian survival converts intersection with slabs into an event to which Gaussian correlation applies. The two Brownian facts below will also be used in the remaining analytic arguments.

Lemma 2.4 (Brownian survival rate). *Let $D \subset \mathbb{R}^m$ be bounded, nonempty, open and convex, and let Q be positive definite and diagonal. If W is Brownian motion with covariance rate $2Q$, started at $x \in D$, then*

$$\lim_{T \rightarrow \infty} -\frac{1}{T} \log \mathbb{P}_x \{W(s) \in D \text{ for all } 0 \leq s \leq T\} = \lambda_Q(D). \quad (16)$$

In particular, the starting point may be 0 when D is centrally symmetric.

Proof. The generator of W is $\text{tr}(Q\nabla^2)$. Let $\tau_D = \inf\{s \geq 0 : W(s) \notin D\}$, and let

$$P_s^D h(x) = \mathbb{E}_x[h(W(s)) \mathbf{1}_{\{s < \tau_D\}}]$$

be its killed semigroup. The corresponding nonnegative self-adjoint operator is the Dirichlet realization of $-\text{tr}(Q\nabla^2)$, with form $\int_D \nabla g^T Q \nabla g$ on $H_0^1(D)$. These standard Dirichlet facts apply here because bounded convex domains are connected and Lipschitz; see, for example, [14, Section 2 and Appendix C], after the linear change of coordinates for Q and the time normalization above. The killed representation can also be obtained first on smooth interior domains and then by exhaustion:

their compactly supported smooth functions exhaust the form domain, while any continuous path remaining in D on a compact time interval has compact image in D .

Write $\lambda = \lambda_Q(D)$. Compactness of $H_0^1(D) \hookrightarrow L^2(D)$ and the spectral theorem give $\|P_s^D\|_{2 \rightarrow 2} = e^{-s\lambda}$ and a first eigenfunction φ that can be chosen nonnegative by taking the absolute value of a Rayleigh minimizer. Interior elliptic regularity and the strong maximum principle make φ strictly positive at every point of D . It is bounded: free heat kernel domination and $\varphi = e^\lambda P_1^D \varphi$ give this directly by Cauchy–Schwarz. Consequently,

$$e^{-T\lambda} \varphi(x) = P_T^D \varphi(x) \leq \|\varphi\|_\infty \mathbb{P}_x\{\tau_D > T\}.$$

For the reverse bound, let p_Q denote the free heat kernel. For $T \geq 1$, the semigroup property and free kernel domination yield

$$\begin{aligned} \mathbb{P}_x\{\tau_D > T\} &\leq \|p_Q(1, x, \cdot)\|_2 \|P_{T-1}^D \mathbf{1}_D\|_2 \\ &\leq \|p_Q(1, x, \cdot)\|_2 |D|^{1/2} e^{-(T-1)\lambda}. \end{aligned}$$

Both prefactors are finite for the fixed D, Q, x . Taking logarithms and dividing by T proves (16). Finally, a nonempty open convex set symmetric about 0 contains 0 as an interior point. $\square$

We will use the following path form of Gaussian correlation. Its finite-grid proof also records how closed constraints are handled.

Lemma 2.5 (Gaussian correlation for path constraints). *Let $X : [0, T] \rightarrow \mathbb{R}^m$ be a centered Gaussian process with continuous paths. For $i = 1, \dots, r$, let E_i be an event of the form*

$$E_i = \bigcap_{j=1}^{r_i} \{A_{ij} X(s) \in K_{ij} \text{ for all } s \in I_{ij}\},$$

where A_{ij} is linear, K_{ij} is a closed centrally symmetric convex set in its target space, and $I_{ij} \subset [0, T]$ is a closed interval. Then

$$\mathbb{P}\left(\bigcap_{i=1}^r E_i\right) \geq \prod_{i=1}^r \mathbb{P}(E_i).$$

Proof. Replace each interval by a finite grid. All the constraints are then closed centrally symmetric convex conditions on one finite-dimensional centered Gaussian vector. Represent that vector as a linear image of a standard Gaussian on its linear support; thus a possibly singular covariance creates no difficulty. Royen’s Gaussian correlation inequality [20], iterated over the finitely many events, gives the desired product bound on the grids. Now refine nested grids whose unions are dense in the respective intervals. Closedness of each K_{ij} and path continuity identify the decreasing limit with the stated event. Continuity of probability from above proves the assertion. $\square$

Proof of Lemma 2.3. The set K is again a bounded open centrally symmetric convex neighborhood of 0. Take Brownian motion W of covariance rate $2Q$ started at 0. On $[0, T]$, impose the closed constraints

$$W(s) \in \overline{D_* D}, \quad |m_i W(s)| \leq \delta/2 \quad (1 \leq i \leq n).$$

Their simultaneous occurrence implies survival in K : $\overline{D_* D} \subset 2D_* D$, since 0 is interior to the convex set D , and the half-threshold slabs lie inside the open full-threshold slabs. By Lemma 2.5, and then by replacing each closed-event probability by its smaller open-event probability,

$$\begin{aligned} &\mathbb{P}_0\{W([0, T]) \subset K\} \\ &\geq \mathbb{P}_0\{W([0, T]) \subset D_* D\} \prod_{i=1}^n \mathbb{P}_0\{|m_i W(s)| < \delta/2 \text{ for all } 0 \leq s \leq T\}. \end{aligned}$$

For a nonzero row m_i , the process $m_i W$ is one-dimensional Brownian motion with covariance rate $2c_i$, where $c_i = m_i Q m_i^T > 0$. The first Dirichlet eigenfunction of $-c_i d^2/dx^2$ on $(-\delta/2, \delta/2)$ is $\cos(\pi x/\delta)$, with eigenvalue $\pi^2 c_i/\delta^2$. Lemma 2.4 gives the corresponding survival exponent. A zero row gives the identically zero process, so its probability is one and its cost is zero.

Take negative logarithms, divide by T , and let $T \rightarrow \infty$. Lemma 2.4 applies to K and $D_* D$, and the change of variables in the Rayleigh quotient gives $\lambda_Q(D_* D) = D_*^{-2} \lambda_Q(D)$. This proves (6). The argument used only inclusions between closed and open events; it requires no assertion about equality of their survival rates. $\square$

It remains to prove the two analytic inputs. We first construct the domain transformations behind Proposition 2.1, and then establish the uniform energy bound in Proposition 2.2.

3 From low energy to shifted signs

We prove Proposition 2.1. Its spectral hypothesis will first give a single normalized function with small derivatives in every coordinate. A lifted Steiner symmetrization then produces a new domain from which one can take one of two signed steps, even after a prescribed small shift. Repeating this construction backwards will allow us to choose the signs forwards. The construction adapts the common-density and prescribed-section arguments of Guo, Fang and Lu [17, Proposition 2.2, Lemmas 3.3–3.4 and 4.1–4.3]. Quadratic-energy versions of this approach appear in Bandeira [5] and Akbas and Sra [1, Lemmas 2.2–2.3]. Here we retain only coordinate Sobolev energies and enough geometric slack for bounded adaptive shifts. The coordinate-energy and shifted-step arguments needed here are proved in full below.

Throughout this section,

$$\kappa = \frac{1}{100}, \quad \delta = \frac{1}{4}, \quad \ell = \frac{5}{2}.$$

3.1 One function for all coordinate energies

We first record a useful boundary convention. If a bounded open convex domain $D \subset \mathbb{R}^m$ contains zero, then

$$H_0^1(D) = \{f|_D : f \in H^1(\mathbb{R}^m), f = 0 \text{ a.e. outside } D\}, \quad (17)$$

where functions on the left are extended by zero. Indeed, for a function on the right, $f_r(x) = f(x/r)$ converges to f in $H^1(\mathbb{R}^m)$ as $r \uparrow 1$. For $r < 1$ its support lies in the compact subset $r\bar{D}$ of D . Mollification with a sufficiently small radius therefore approximates f_r by functions in $C_c^\infty(D)$. Conversely, zero extension preserves the H^1 norm on $C_c^\infty(D)$ and hence on its completion.

Lemma 3.1. *Let $K \subset \mathbb{R}^m$ be a nonempty bounded open convex set symmetric about zero. If*

$$\lambda_Q(K) \leq \kappa^2 \operatorname{tr} Q \quad \text{for every positive definite diagonal } Q,$$

then there is a nonnegative $f \in H^1(\mathbb{R}^m)$, vanishing almost everywhere outside K , such that

$$\int f^2 = 1, \quad \int |\partial_i f|^2 \leq \kappa^2 \quad (1 \leq i \leq m).$$

Proof. Let $\mathcal{F}$ be the nonnegative functions in $H^1(\mathbb{R}^m)$ that vanish almost everywhere outside K and have L^2 norm one, and put $E_i(f) = \int |\partial_i f|^2$. Consider the set of simultaneous upper bounds

$$\mathcal{A} = \{u \in \mathbb{R}^m : E_i(f) \leq u_i \text{ for every } i \text{ for some } f \in \mathcal{F}\}.$$

It is nonempty and is closed under addition of nonnegative vectors. It is also convex. For $f, g \in \mathcal{F}$ and $0 \leq t \leq 1$, the function $h = (tf^2 + (1-t)g^2)^{1/2}$ belongs to $\mathcal{F}$ and satisfies

$$|\partial_i h|^2 \leq t|\partial_i f|^2 + (1-t)|\partial_i g|^2 \quad \text{a.e.}$$

This follows from the Sobolev chain rule for the Euclidean norm and Cauchy–Schwarz.

To see that $\mathcal{A}$ is closed, let $u^{(j)} \in \mathcal{A}$ converge to u , and choose corresponding $f_j \in \mathcal{F}$. Their H^1 norms are bounded. Since all supports lie in a fixed bounded set, weak H^1 compactness and compactness in L^2 give a subsequence converging weakly in H^1 and strongly in L^2 to a member f of $\mathcal{F}$. Lower semicontinuity gives $E_i(f) \leq u_i$ for every i .

If $c = \kappa^2(1, \dots, 1)$ did not belong to $\mathcal{A}$, strict separation would give a nonzero $w \in \mathbb{R}^m$ with

$$w \cdot c < \inf_{u \in \mathcal{A}} w \cdot u.$$

The upper-set property forces every w_i to be nonnegative. For $\epsilon > 0$ let $Q_\epsilon = \text{diag}(w_i + \epsilon)$. By (17), taking absolute values and normalizing does not change the Rayleigh infimum, so

$$\inf_{u \in \mathcal{A}} w \cdot u = \inf_{f \in \mathcal{F}} \sum_i w_i E_i(f) \leq \lambda_{Q_\epsilon}(K) \leq \kappa^2 \sum_i (w_i + \epsilon).$$

Letting $\epsilon \downarrow 0$ contradicts separation. This argument also covers zero components of w . Thus $c \in \mathcal{A}$, as required. $\square$

3.2 Convexity under Minkowski averages

We will apply Jensen’s inequality to the energies of horizontal sections. The needed property is a diagonal version of the classical Brascamp–Lieb convexity inequality for the first Dirichlet eigenvalue [10, Theorem 6.2]; see also the statement in Bryan, Clutterbuck and Rankin [11, Theorem 1.2]. We give the proof from Gaussian log-concavity and the Brownian survival rate, including the open-domain approximation used here.

Lemma 3.2. *Let $D_0, D_1 \subset \mathbb{R}^m$ be nonempty bounded open convex sets symmetric about zero, and let Q be positive definite diagonal. For $0 \leq \alpha \leq 1$,*

$$\lambda_Q((1-\alpha)D_0 + \alpha D_1) \leq (1-\alpha)\lambda_Q(D_0) + \alpha\lambda_Q(D_1).$$

Proof. The endpoints are immediate, so suppose $0 < \alpha < 1$ and write $D_\alpha = (1-\alpha)D_0 + \alpha D_1$. Fix $0 < \eta < 1$, set $r = 1 - \eta$, and put

$$A_i = r\overline{D_i}, \quad A_\alpha = (1-\alpha)A_0 + \alpha A_1.$$

These are compact convex sets, and $A_\alpha \subset D_\alpha$: indeed, $A_\alpha \subset r\overline{D_\alpha}$, which is compactly contained in D_α .

Let W start at zero with Brownian covariance rate $2Q$. On a finite grid of positive times in $[0, T]$, its values form a Gaussian vector. Gaussian log-concavity [19, Theorem 2] shows that

$$\mathbb{P}\{W \text{ lies in } A_\alpha \text{ on the grid}\} \geq \prod_{i=0}^1 \mathbb{P}\{W \text{ lies in } A_i \text{ on the grid}\}^{\alpha_i}, \quad (\alpha_0, \alpha_1) = (1-\alpha, \alpha).$$

Here the Minkowski average of the two grid events is contained in the grid event for A_α . Increase the grids to a dense subset of $[0, T]$. Closedness of the A_i and continuity of Brownian paths allow passage to the decreasing limits of these events. Since $rD_i \subset A_i$ and $A_\alpha \subset D_\alpha$, it follows that

$$\mathbb{P}_0\{W([0, T]) \subset D_\alpha\} \geq \prod_{i=0}^1 \mathbb{P}_0\{W([0, T]) \subset rD_i\}^{\alpha_i}.$$

Lemma 2.4, followed by the scaling of the Dirichlet quotient, now gives

$$\lambda_Q(D_\alpha) \leq r^{-2}((1-\alpha)\lambda_Q(D_0) + \alpha\lambda_Q(D_1)).$$

Finally let $\eta \downarrow 0$. □

3.3 A section that permits shifted signed steps

The next lemma is the geometric step in the signing construction. Its conclusion holds for every allowed shift after the point in the new domain has been chosen.

Lemma 3.3. *Let $K \subset \mathbb{R}^m$ be a nonempty bounded open convex set symmetric about zero and satisfying $\lambda_Q(K) \leq \kappa^2 \operatorname{tr} Q$ for every positive definite diagonal Q . For every coordinate vector e , there is another domain $T_e(K)$ with the same properties such that, for all $y \in T_e(K)$ and $|q| \leq \delta$, some $\varepsilon \in \{-1, 1\}$ satisfies*

$$y + (q + \varepsilon)e \in K.$$

Proof. Lift K to the bounded open convex set

$$B = \{(y, s) \in \mathbb{R}^m \times \mathbb{R} : y + se \in K, |s| < \ell\}.$$

Let P be its projection onto $\mathbb{R}^m$. For $y \in P$, its vertical fiber $I_y = \{s : (y, s) \in B\}$ is a nonempty open interval. Write $L(y) = |I_y|$. Convexity gives

$$(1-t)I_y + tI_z \subset I_{(1-t)y + tz}, \quad L((1-t)y + tz) \geq (1-t)L(y) + tL(z).$$

Thus L is positive, concave and continuous on the open convex set P . The vertical Steiner symmetrization of B is

$$B^* = \{(y, s) : y \in P, |s| < L(y)/2\}.$$

This set is bounded, open and convex. Symmetry of K implies $L(-y) = L(y)$, so B^* is symmetric separately in y and in s . For $s \geq 0$ put

$$D_s = \{y \in P : L(y) > 2s\}, \quad b = \sup\{s \geq 0 : D_s \neq \emptyset\}.$$

Then $0 < b \leq \ell$. Each D_s with $0 \leq s < b$ is a nonempty bounded open convex set symmetric about zero, and these domains decrease as s increases.

The section D_1 is the target because each of its points permits the required shifted signed step. Indeed, for any $y \in D_1$, write $I_y = (a, b_0)$. Its length exceeds two. Choose $c \in (a+1, b_0-1)$; then

$$[c-1, c+1] \subset I_y, \quad |c| < \ell-1 = \frac{3}{2}.$$

For $|q| \leq \delta$, we have $|c-q| < 7/4 < 2$. If $q \leq c$, then $q+1$ lies in $[c-1, c+1]$; if $q \geq c$, then $q-1$ lies in that segment. Thus one of $q+1, q-1$ lies in the original open fiber, giving membership in K itself. It remains to prove that D_1 is nonempty and has the same spectral bounds as K .

Choose f from Lemma 3.1, and define

$$\Psi(y, s) = (2\ell)^{-1/2} f(y + se) \mathbf{1}_{(-\ell, \ell)}(s).$$

Translation invariance and Fubini give

$$\|\Psi\|_2 = 1, \quad \int |\partial_{y_i} \Psi|^2 = \int |\partial_i f|^2 \leq \kappa^2. \quad (18)$$

Only derivatives in the horizontal variable y will be used.

Rearrange each nonnegative fiber $\Psi(y, \cdot)$ into an even function nonincreasing in $|s|$, denoted $\Psi^*(y, \cdot)$. A jointly measurable version is obtained from the measurable superlevel lengths $\mu(y, u) = |\{s : \Psi(y, s) > u\}|$ by replacing each superlevel set with its centered interval of length $\mu(y, u)$. This preserves the L^2 norm. Since Ψ vanishes almost everywhere outside B , its rearrangement vanishes almost everywhere outside B^* .

For nonnegative $a, b \in L^2(\mathbb{R})$, the layer-cake formula gives

$$\int ab = \int_0^\infty \int_0^\infty |\{a > u\} \cap \{b > v\}| du dv \leq \int a^* b^*.$$

The intersection is bounded by the smaller of the two superlevel measures, and centered intervals attain that bound. Since rearrangement preserves the individual norms, $\|a^* - b^*\|_2 \leq \|a - b\|_2$. Applying this to the fibers at y and $y + he_i$, then integrating over y , bounds the horizontal difference quotients of Ψ^* by those of Ψ . Their weak limits yield

$$\int |\partial_{y_i} \Psi^*|^2 \leq \int |\partial_i f|^2 \leq \kappa^2. \quad (19)$$

Sobolev slicing and Fubini therefore imply that, for almost every s , $\Psi^*(\cdot, s) \in H^1(\mathbb{R}^m)$ and vanishes almost everywhere outside $D_{|s|}$. Each nonzero such slice belongs to $H_0^1(D_{|s|})$ by (17).

We next show that the rearranged density puts enough mass at heights above one. Its mean absolute height is

$$\begin{aligned} M &= \iint |s| \Psi^*(y, s)^2 dy ds \\ &= \frac{1}{8\ell} \int_{-\ell}^\ell \int_{-\ell}^\ell \int_{\mathbb{R}^m} \min\{f(y + re)^2, f(y + r'e)^2\} dy dr dr'. \end{aligned} \quad (20)$$

Indeed, a centered interval of length u has integral of $|s|$ equal to $u^2/4$. Apply this to the superlevel intervals of Ψ^{*2} and use Tonelli; the factor $(2\ell)^{-1}$ in Ψ^2 gives the displayed normalization. For nonnegative translates u, v of f , both of norm one,

$$\int \min(u^2, v^2) = 1 - \frac{1}{2} \|u^2 - v^2\|_1 \geq 1 - \|u - v\|_2.$$

Moreover,

$$\|f(\cdot + re) - f(\cdot + r'e)\|_2 \leq |r - r'| \|\partial_e f\|_2 \leq 2\ell\kappa.$$

It follows from (20) that

$$M \geq \frac{\ell}{2} (1 - 2\ell\kappa) = \frac{19}{16} > 1. \quad (21)$$

Under the probability density $\Psi^{*2}(y, s)$, we have $|s| < b$ almost surely; hence $M < b$. In particular, D_1 is nonempty.

It remains to pass the derivative bounds to this particular section. Fix a positive definite diagonal Q and put $h_Q(s) = \lambda_Q(D_s)$ for $0 \leq s < b$. This function is finite and nondecreasing. Convexity of B^* , domain monotonicity and Lemma 3.2 give

$$\begin{aligned} h_Q((1-t)s + tr) &\leq \lambda_Q((1-t)D_s + tD_r) \\ &\leq (1-t)h_Q(s) + th_Q(r), \end{aligned}$$

so h_Q is convex. Give s the probability density $\rho(s) = \int \Psi^*(y, s)^2 dy$. Testing the Rayleigh quotient with each nonzero slice and then integrating, we obtain

$$\int h_Q(|s|)\rho(s) ds \leq \iint \langle Q \nabla_y \Psi^*, \nabla_y \Psi^* \rangle \leq \kappa^2 \operatorname{tr} Q.$$

This also proves the integrability needed for Jensen's inequality. Since $1 < M < b$, monotonicity and Jensen give

$$\lambda_Q(D_1) = h_Q(1) \leq h_Q(M) \leq \int h_Q(|s|)\rho(s) ds \leq \kappa^2 \operatorname{tr} Q.$$

The construction of D_1 is independent of Q , so every required spectral bound is preserved. Taking $T_e(K) = D_1$ proves the lemma. $\square$

3.4 Choosing the signs

Proof of Proposition 2.1. Write the prescribed coordinate directions as $e_{j_1}, \dots, e_{j_s}$. Construct domains backwards by

$$K_s = K, \quad K_{i-1} = T_{e_{j_i}}(K_i) \quad (i = s, s-1, \dots, 1).$$

Lemma 3.3 preserves all domain and spectral hypotheses. Each K_i therefore contains zero, and we may start with $x^{(0)} = 0 \in K_0$. Suppose $x^{(i-1)} \in K_{i-1}$, and set

$$q = q_i(x_{j_1}^{(i-1)}, \dots, x_{j_{i-1}}^{(i-1)}).$$

Since $q \in [-\delta, \delta]$, the lemma supplies $\varepsilon_i \in \{-1, 1\}$ with

$$x^{(i)} = x^{(i-1)} + (q + \varepsilon_i)e_{j_i} \in K_i.$$

Induction gives $x^{(s)} \in K_s = K$ and the required signs. Distinctness of the coordinate directions ensures that later steps do not alter the preceding coefficient coordinates on which a shift depends. The backwards domains may depend on the entire input sequence, as allowed in the proposition. $\square$

4 Matrix variation along the filter

The proof of Proposition 2.2 must control all the states of a filtered Gaussian vector, even when the sequence is arbitrarily long. We first prove a deterministic estimate that measures variation of the filter by a total energy, rather than by the number of steps. Spectral localization will give separate bounds for changes in the sequence index and in Ornstein–Uhlenbeck time.

Throughout this section, fix a recursion as in Proposition 2.2 and a positive definite diagonal matrix $Q \in \mathbb{R}^{m \times m}$. The sequence index is $t \in \{0, \dots, n\}$. For a real matrix A , write $\|A\|_{\text{HS}}^2 = \operatorname{tr}(AA^\top)$.

4.1 Trace budgets and spectral scales

The support of R_{t-1} is contained in the initial block and the first $t-1$ new coordinates. Hence the new coordinate has no cross term in $R_t R_t^\top$, and induction gives

$$R_t R_t^\top = I_d, \quad L_t := R_t Q R_t^\top = C_t L_{t-1} C_t^\top + Q_{d+t, d+t} b_t b_t^\top. \quad (22)$$

In particular L_t is positive definite. Every C_t is a contraction, since $C_t C_t^\top \preceq I_d$.

Let ℓ_t be the trace lost in the contraction in (22), and let g_t be the trace added afterward. Both are nonnegative. Since $\|b_t\| \leq 1$,

$$\sum_{t=1}^n g_t \leq \sum_{t=1}^n Q_{d+t,d+t}, \quad \sum_{t=1}^n \ell_t = \text{tr } L_0 + \sum_{t=1}^n g_t - \text{tr } L_n \leq \text{tr } Q. \quad (23)$$

These two budgets are independent of the length of the sequence.

Fix the following piecewise linear cutoff on $(0, \infty)$:

$$f_0(x) = \begin{cases} 0, & 0 < x \leq \frac{1}{2}, \\ 2x - 1, & \frac{1}{2} < x < 1, \\ 1, & 1 \leq x \leq 2, \\ 2 - \frac{x}{2}, & 2 < x < 4, \\ 0, & x \geq 4. \end{cases}$$

Thus $0 \leq f_0 \leq 1$ and its Lipschitz constant is 2. We use the scalar functions

$$\begin{aligned} a = \frac{3}{2}, \quad w(x) &= \min\{x^a, x^{-1}\}, \quad p(x) = \int_0^x \frac{w(u)}{u} du, \\ p_1(x) &= \int_0^x \frac{u^{a-1}}{(1+u)^a} du, \quad \phi(x) = \min\{x^{a-1}, x^{-1}\}. \end{aligned} \quad (24)$$

The potential p will control changes of the cutoff between endpoint matrices, while p_1 will control gains accumulated through intervening contractions. The function ϕ measures the contribution of each spectral direction to the size of a scale. Matrix functions below are defined by spectral calculus.

For each dyadic scale $\theta \in 2^{\mathbb{Z}}$, set

$$X_t = L_t/\theta, \quad F_t^\theta = R_t^\top f_0(X_t) R_t, \quad k_t(\theta) = \text{tr } \phi(X_t).$$

Here R_t is $d \times m$, X_t is $d \times d$, and F_t^θ is $m \times m$. By (22), $R_t^\top$ is an isometry, so the lift preserves the localized state norm:

$$\|F_t^\theta x\| = \|f_0(X_t) R_t x\| \quad (x \in \mathbb{R}^m).$$

It also removes dependence on state coordinates: replacing R_t by OR_t , with O orthogonal, replaces X_t by $OX_t O^\top$ and leaves F_t^θ unchanged. The notation X_t always refers to the scale currently under consideration; at a fixed scale we also abbreviate $F_t = F_t^\theta$ and $k_t = k_t(\theta)$. The geometric decay of ϕ at zero and infinity gives

$$\sum_{\theta \in 2^{\mathbb{Z}}} k_t(\theta) \leq Cd. \quad (25)$$

Indeed, for each positive eigenvalue λ of L_t , the terms $\phi(\lambda/\theta)$ on either side of $\theta = \lambda$ form bounded geometric sums. On the support band of f_0 , one has $\phi \geq 1/4$. It follows that

$$\begin{aligned} \text{tr } f_0(X_t)^2 &\leq 4k_t, \quad \text{tr}(X_t f_0(X_t)^2) \leq 16k_t, \quad \|F_t\|_{\text{op}} \leq 1, \\ F_t \neq 0 &\implies k_t \geq \frac{1}{4}. \end{aligned} \quad (26)$$

Every eigenvalue of L_t lies between the smallest and largest diagonal entries of Q . Consequently only finitely many scales have $F_t^\theta \neq 0$ for some t , and every such scale satisfies $\theta \leq 2\|Q\|_{\text{op}}$.

We now assign an additive energy to intervals of sequence indices. Put $h = p + p_1$ and $\hat{X}_i = C_i X_{i-1} C_i^\top$. Define

$$E_\theta(s, t) = \sum_{i=s+1}^t \left\{ \text{tr } h(X_{i-1}) - \text{tr } h(\hat{X}_i) + \text{tr } h(X_i) - \text{tr } h(\hat{X}_i) \right\}, \quad 0 \leq s \leq t \leq n. \quad (27)$$

The ordered eigenvalues decrease under a contraction and increase under a positive semidefinite gain. Since h is increasing, every trace difference in braces is nonnegative. In particular, E_θ is nonnegative and additive on adjacent index intervals.

The derivatives of the two functions satisfy, uniformly for $x > 0$,

$$\sum_{\theta \in 2^{\mathbb{Z}}} (p'(x/\theta) + p'_1(x/\theta)) \leq C.$$

For p' , use the bounds $u^{1/2}$ for $u \leq 1$ and u^{-2} for $u \geq 1$; for p'_1 , use $u^{1/2}$ and u^{-1} respectively. All four dyadic sums are geometric. Pairing ordered eigenvalues before and after each contraction or gain, and integrating the last bound between each pair, proves

$$\sum_{\theta \in 2^{\mathbb{Z}}} \theta E_\theta(0, n) \leq C \left(\sum_i \ell_i + \sum_i g_i \right) \leq C \operatorname{tr} Q. \quad (28)$$

The factor θ cancels the factor $1/\theta$ from differentiating $p(x/\theta)$ or $p_1(x/\theta)$.

Here is the variation estimate to be used in the next section.

Lemma 4.1 (Weighted increment estimate). *There are absolute constants $e_0 > 0$ and $C < \infty$ with the following property. For every recursion in Proposition 2.2, every positive definite diagonal Q , every dyadic θ , and $0 \leq s \leq t \leq n$, if $E = E_\theta(s, t) \leq e_0$, then*

$$\|(F_t^\theta - F_s^\theta)(I_m + Q/\theta)^{1/2}\|_{\text{HS}}^2 \leq C(1 + k_t(\theta))E^{1/a}, \quad a = \frac{3}{2}. \quad (29)$$

We prove this in three steps. A scalar inequality first controls changes of spectral cutoffs. Exact matrix expansions then compare the two endpoints using the old part of the state and the accumulated new gains. Finally, a separate estimate bounds those accumulated gains by the additive energy, even through noncommuting contractions.

4.2 A scalar comparison

For $x, y > 0$, define

$$\mathcal{J}(x, y) = p(x) - p(y) + \left(\frac{y}{x} - 1 \right) \frac{w(x) + w(y)}{2}.$$

Lemma 4.2. *There is an absolute $c > 0$ such that, for all $x, y > 0$,*

$$\begin{aligned} \mathcal{J}(x, y) &\geq c(1 + y)|f_0(x) - f_0(y)|^2, \\ \frac{y}{x}\mathcal{J}(y, x) &\geq c(1 + y)|f_0(x) - f_0(y)|^2. \end{aligned} \quad (30)$$

Proof. The function $u \mapsto \log w(e^u) = \min\{au, -u\}$ is a -Lipschitz. Write $r = y/x$. For $r < 1$, the two endpoint lower bounds give

$$w(u) \geq \max\{w(x)(u/x)^a, w(y)(y/u)^a\}, \quad y \leq u \leq x.$$

The two expressions cross inside $[y, x]$, by the same Lipschitz bound. Integrating them against du/u yields

$$p(x) - p(y) \geq \frac{w(x) + w(y) - 2\sqrt{w(x)w(y)}r^{a/2}}{a}.$$

For $r > 1$, integration of the minimum of the corresponding endpoint upper bounds on $[x, y]$ gives

$$p(y) - p(x) \leq \frac{2\sqrt{w(x)w(y)}r^{a/2} - w(x) - w(y)}{a}.$$

Using $2\sqrt{w(x)w(y)} \leq w(x) + w(y)$ in either case, we obtain

$$\mathcal{J}(x, y) \geq \frac{w(x) + w(y)}{2}B(r), \quad B(r) = r - 1 - \frac{r^{a/2} - 1}{a/2}. \quad (31)$$

If $b = a/2 = 3/4$, then $B(1) = B'(1) = 0$ and $B''(r) = (1-b)r^{b-2} > 0$. Moreover $B(0+) = 1/b - 1 > 0$ and $B(r)/r \rightarrow 1$ at infinity. Thus, with an absolute positive constant,

$$B(r) \geq c \begin{cases} 1, & 0 < r \leq \frac{1}{2}, \\ (r-1)^2, & \frac{1}{2} \leq r \leq 2, \\ r, & r \geq 2. \end{cases}$$

The function $rB(1/r)$ satisfies the same bounds, after changing c . Applying (31) with the arguments reversed therefore gives these bounds for both left sides of (30).

If $f_0(x) \neq f_0(y)$, at least one argument belongs to $[1/2, 4]$. When $1/2 \leq r \leq 2$, both arguments lie in $[1/4, 8]$, where w is bounded below and $1+y$ is bounded above. The Lipschitz bound $|f_0(x) - f_0(y)| \leq 2x|r-1|$ applies. When $r \leq 1/2$, one w value is bounded below and $y \leq 4$. When $r \geq 2$, one w value is again bounded below, and either $y \leq 4$ or $y = rx \leq 4r$. The preceding three bounds, together with $|f_0(x) - f_0(y)| \leq 1$, prove the result. If the cutoff values are equal, the right sides vanish and the result follows directly from (31). $\square$

4.3 Comparing the endpoints

Fix a scale and indices $s \leq t$. Abbreviate

$$Y = X_s, \quad X = X_t, \quad \mathcal{C} = C_t \cdots C_{s+1}, \quad Z = CYC^\top, \quad V = X - Z.$$

An empty product is the identity. The recursion shows that V is positive semidefinite: it is the sum of all intervening gains, each transported through the contractions that follow it. The matrices Y, X, Z are positive definite, since every C_i is invertible.

Lemma 4.3 (Endpoint comparison). *With the preceding notation,*

$$\|(F_t - F_s)(I_m + Q/\theta)^{1/2}\|_{\text{HS}}^2 \leq C \{ \text{tr}(p(Y) - p(Z)) + \text{tr}(p(X) - p(Z)) \}. \quad (32)$$

Proof. Let P_s be the coordinate projection onto the initial block and the first s new coordinates, and put

$$Q_{\text{old}} = P_s(Q/\theta)P_s, \quad Q_{\text{new}} = Q/\theta - Q_{\text{old}}.$$

Diagonality of Q makes these nonnegative matrices with disjoint coordinate supports. Since $R_t P_s = C R_s$, the support identities are

$$\begin{aligned} R_t R_s^\top &= \mathcal{C}, & R_t Q_{\text{old}} R_s^\top &= CY, & R_t Q_{\text{old}} R_t^\top &= Z, \\ R_t Q_{\text{new}} R_t^\top &= V, & F_s Q_{\text{new}} &= 0. \end{aligned} \quad (33)$$

We split the right weight into $I_m + Q_{\text{old}}$ and Q_{new} , and use the intermediate matrix $\tilde{F} = R_t^\top f_0(Z) R_t$ for the first part.

The old state under contraction. Choose orthonormal eigenbases of Z and Y , with eigenvalues z_i and y_j . Let m_{ij} be the squared entries of $\mathcal{C}$ in these bases. Its row and column sums are at most one; denote their deficits by

$$d_i = 1 - \sum_j m_{ij}, \quad h_j = 1 - \sum_i m_{ij}.$$

The matrix $n_{ij} = m_{ij}y_j/z_i$ has both row and column sums one. For rows this follows from $Z = \mathcal{C}Y\mathcal{C}^\top$; for columns it follows from the identity

$$\mathcal{C}^\top Z^{-1} \mathcal{C} = Y^{-1},$$

which uses invertibility of $\mathcal{C}$.

Expanding the weighted squared distance with (33), its two diagonal terms are $\text{tr}(f_0(Z)^2(I_d + Z))$ and $\text{tr}(f_0(Y)^2(I_d + Y))$. The cross term, before multiplication by two, is $\sum_{ij} m_{ij}(1 + y_j)f_0(z_i)f_0(y_j)$. Since $\sum_j m_{ij}y_j = z_i$, this gives the exact expansion

$$\begin{aligned} & \|(\tilde{F} - F_s)(I_m + Q_{\text{old}})^{1/2}\|_{\text{HS}}^2 \\ &= \sum_{ij} m_{ij}(1 + y_j)(f_0(z_i) - f_0(y_j))^2 + \sum_i d_i f_0(z_i)^2 + \sum_j h_j(1 + y_j)f_0(y_j)^2. \end{aligned} \quad (34)$$

Expanding $\mathcal{J}$ instead, the two stochastic sums of n and the relation $n_{ij}z_i/y_j = m_{ij}$ yield

$$\text{tr}(p(Y) - p(Z)) = \sum_{ij} n_{ij} \mathcal{J}(y_j, z_i) + \frac{1}{2} \sum_i d_i w(z_i) + \frac{1}{2} \sum_j h_j w(y_j). \quad (35)$$

The second inequality of Lemma 4.2 controls the double sum in (34). The remaining terms are controlled by the deficits in (35), since both $f_0(u)^2$ and $(1 + u)f_0(u)^2$ are bounded by an absolute multiple of $w(u)$. This proves the required bound for $\tilde{F} - F_s$.

The accumulated gain. Now choose eigenbases of X and Z , with eigenvalues x_i and z_j , and let m_{ij} be the squared entries of their orthogonal change of basis. Both sums of m equal one. Set $n_{ij} = m_{ij}z_j/x_i$. Its row sums are at most one by $Z \preceq X$, and its column sums are at most one by $X^{-1} \preceq Z^{-1}$: explicitly, for a unit z_j -eigenvector v_j ,

$$\sum_i n_{ij} = z_j v_j^\top X^{-1} v_j \leq z_j v_j^\top Z^{-1} v_j = 1.$$

Write d'_i, h'_j for these row and column deficits. Expansion of $\mathcal{J}$ now gives

$$\text{tr}(p(X) - p(Z)) = \sum_{ij} m_{ij} \mathcal{J}(x_i, z_j) + \frac{1}{2} \sum_i d'_i w(x_i) + \frac{1}{2} \sum_j h'_j w(z_j). \quad (36)$$

The coisometry $R_t R_t^\top = I_d$ and (33) show that

$$\begin{aligned} & \|(F_t - \tilde{F})(I_m + Q_{\text{old}})^{1/2}\|_{\text{HS}}^2 \\ &= \|(f_0(X) - f_0(Z))(I_d + Z)^{1/2}\|_{\text{HS}}^2 = \sum_{ij} m_{ij}(1 + z_j)(f_0(x_i) - f_0(z_j))^2. \end{aligned}$$

The first inequality of Lemma 4.2 controls this by (36).

It remains to include the new-coordinate part of the right weight. By (33), it is exactly

$$\|(F_t - F_s)Q_{\text{new}}^{1/2}\|_{\text{HS}}^2 = \text{tr}(f_0(X)^2 V) = \sum_i x_i d'_i f_0(x_i)^2. \quad (37)$$

Here $x_i d'_i$ is the i th diagonal entry of $X - Z$ in the X -eigenbasis. The bound $uf_0(u)^2 \leq Cw(u)$ controls this term by the row deficits in (36). Finally apply the squared triangle inequality to $(F_t - \tilde{F}) + (\tilde{F} - F_s)$ with the old weight, and add (37). This proves (32). $\square$

The endpoint comparison has accounted for all coordinate weights. Its remaining difficulty is that $p(X) - p(Z)$ describes a single accumulated gain, whereas the energy in (27) records gains separated by contractions. The next estimate connects these two quantities.

4.4 Controlling the accumulated gain

Lemma 4.4. *For the matrices X, Y, Z, V above, suppose $E = E_\theta(s, t) \leq e_0$, where*

$$c_a = a2^a, \quad e_0 = \frac{1}{16c_a}, \quad a = \frac{3}{2}.$$

If $q = (c_a E)^{1/a}$, then $q < 1/4$ and

$$V \preceq q(I_d + X), \quad \text{tr}(p(X) - p(Z)) \leq Cq \text{tr} \phi(X). \quad (38)$$

Proof. Follow the old and new parts separately through the interval $[s, t]$. Start with $(Z_*, V_*) = (Y, 0)$. At each contraction, replace both matrices by their congruences under C_i . At each gain, add the positive semidefinite gain to V_* only, interpolating linearly during this addition. At every stage $D = Z_* + V_*$ is the matrix in the original contraction-and-gain path; at the endpoint $(Z_*, V_*) = (Z, V)$.

Define

$$S = I_d + Z_*, \quad A = S^{-1/2} V_* S^{-1/2}, \quad G = \text{tr} A^a.$$

The use of $I_d + Z_*$ makes this quantity stable under contractions. We will show that its increases during gains are paid for by the p_1 part of the energy.

Contractions do not increase G . For a contraction C , put $S' = I_d + CZ_*C^\top$ and $T_c = (S')^{-1/2}CS^{1/2}$. The new relative matrix is $A' = T_c A T_c^\top$. Moreover

$$S' - CSC^\top = I_d - CC^\top \succeq 0, \quad T_c T_c^\top \preceq I_d.$$

Thus the ordered eigenvalues of A' are no larger than those of A ; for example, compare the singular values of $T_c A^{1/2}$ and $A^{1/2}$. Raising these nonnegative eigenvalues to the power a and summing proves $\text{tr}(A')^a \leq \text{tr} A^a$. This argument uses eigenvalue comparison, not operator monotonicity of the power a .

The derivative during a gain. Here S is fixed and $\dot{V}_* \succeq 0$. Introduce

$$J = I_d + D = S + V_*, \quad H = J^{-1/2} V_* J^{-1/2}, \quad T = S^{-1/2} J^{1/2}.$$

Since $I_d - H = J^{-1/2} S J^{-1/2}$ is positive definite,

$$T^\top T = (I_d - H)^{-1}.$$

Write the polar decomposition as $T = U(I_d - H)^{-1/2}$, where U is orthogonal. Then

$$A = THT^\top = U[H(I_d - H)^{-1}]U^\top. \quad (39)$$

In particular, the eigenvalues of A are $h/(1 - h)$, where h ranges over the eigenvalues of H .

For a continuously differentiable scalar function g on the spectral range of a differentiable symmetric matrix path $A(\xi)$, $\frac{d}{d\xi} \text{tr} g(A) = \text{tr}(g'(A)\dot{A})$. One can verify this first for polynomials by cyclicity of the trace, then approximate g' uniformly by polynomials and integrate along the path. Applied to $g(u) = u^a$, this gives

$$\dot{G} = a \text{tr}(S^{-1/2}A^{a-1}S^{-1/2}\dot{V}_*). \quad (40)$$

This formula remains valid when A is singular, because $u \mapsto u^a$ is continuously differentiable at zero for $a > 1$. The coefficient in this trace has the exact conjugation identity

$$\begin{aligned} J^{1/2}S^{-1/2}A^{a-1}S^{-1/2}J^{1/2} &= T^\top A^{a-1}T \\ &= H^{a-1}(I_d - H)^{-a}. \end{aligned} \quad (41)$$

The second equality follows from (39) and the polar decomposition. The factors on its right commute because they are functions of H ; no commutation of Z_* and V_* is assumed.

As long as $G \leq 1$, every eigenvalue of A is at most one, so (39) gives $H \preceq I_d/2$. Since $a - 1 = 1/2$, spectral calculus in H then gives

$$H^{a-1}(I_d - H)^{-a} \preceq 2^a H^{1/2}.$$

Also $V_* \preceq D$, and hence

$$H \preceq J^{-1/2}DJ^{-1/2} = D(I_d + D)^{-1}.$$

The square root preserves positive semidefinite order. One direct proof is the convergent integral formula

$$M^{1/2} = \frac{1}{\pi} \int_0^\infty u^{-1/2} M(uI_d + M)^{-1} du, \quad M \succeq 0 :$$

the integrand preserves order because $M(uI_d + M)^{-1} = I_d - u(uI_d + M)^{-1}$ and inversion reverses order. It follows that

$$H^{1/2} \preceq D^{1/2}(I_d + D)^{-1/2}.$$

Conjugating (41) back by $J^{-1/2}$ yields

$$S^{-1/2}A^{a-1}S^{-1/2} \preceq 2^a D^{1/2}(I_d + D)^{-3/2} = 2^a p'_1(D).$$

Pair this inequality with $\dot{V}_* \succeq 0$ in (40). Since $\dot{D} = \dot{V}_*$ during a gain, we obtain the conditional differential bound

$$\dot{G} \leq c_a \text{tr}(p'_1(D)\dot{V}_*) = c_a \frac{d}{d\xi} \text{tr} p_1(D), \quad G \leq 1. \quad (42)$$

Closing the bound. Initially $G = 0$, and contractions cannot increase it. The sum of all increases of $\text{tr} p_1(D)$ during gains is at most E , by (27). If G first reached one during a gain, integration of (42) up to that point would give $G \leq c_a E \leq 1/16$, a contradiction. Thus the differential bound applies throughout, and $G \leq c_a E$ at the endpoint. In particular,

$$A \preceq qI_d, \quad V \preceq q(I_d + Z) \preceq q(I_d + X), \quad q \leq 16^{-2/3} < \frac{1}{4}.$$

If $E = 0$, this already gives $V = 0$ and the second conclusion in (38). Suppose now that $q > 0$.

The ordered eigenvalues of X and Z obey

$$x_j \geq z_j \geq (1 - q)x_j - q,$$

because $X - Z = V \preceq q(I_d + X)$. For a corresponding pair x, z , if $x \leq 4q \leq 1$, then

$$p(x) - p(z) \leq p(x) = \frac{x^a}{a} \leq \frac{4}{a}qx^{a-1} = \frac{4}{a}q\phi(x).$$

If $x > 4q$, then $z \geq x/2$ and $x - z \leq q(1 + x)$. On $[x/2, x]$ we have $w(u) \leq 2^a w(x)$ and $u^{-1} \leq 2/x$, so

$$p(x) - p(z) \leq Cq(1 + x^{-1})w(x) \leq Cq\phi(x).$$

The last inequality holds for both $x \leq 1$ and $x \geq 1$ directly from (24). Summing over the eigenvalues proves the trace bound in (38). $\square$

Proof of Lemma 4.1. The total variation of $\text{tr } p$ along the separate contraction and gain steps from s to t is at most E . Therefore

$$|\text{tr } p(Y) - \text{tr } p(X)| \leq E.$$

Together with Lemma 4.4, this bounds the sum on the right of (32) by

$$E + 2 \text{tr}(p(X) - p(Z)) \leq E + CE^{1/a}k_t.$$

Since $0 \leq E \leq e_0 \leq 1$, we have $E \leq E^{1/a}$. Lemma 4.3 now proves (29), including $E = 0$. $\square$

The two estimates needed for the probability argument are now in place. Equation (28) bounds the sum of the energies over all spectral scales, while Lemma 4.1 turns a short interval in this energy into a small Gaussian increment. In particular, zero energy implies $F_t^\theta = F_s^\theta$ exactly, so repeated indices at zero distance will not increase the size of a covering in the next section.

5 Uniform Gaussian survival

We now prove Proposition 2.2. The matrix estimates of the preceding section measure how much the filtered constraints change as the sequence index varies. At a fixed spectral scale, we use that estimate to control each bounded-energy group of states over one short interval of Ornstein–Uhlenbeck time with a fixed positive probability. Gaussian correlation then combines these intervals and the spectral scales. Their total cost per unit time is bounded by $C \text{tr } Q$, which gives the required Dirichlet-energy bound.

5.1 Stationary survival and Dirichlet energy

Let Q be a positive definite diagonal matrix on $\mathbb{R}^m$, and let $U(\tau)$ be the stationary solution of

$$dU = -QU \, d\tau + \sqrt{2Q} \, dB(\tau).$$

Here B is standard m -dimensional Brownian motion and $U(0)$ has standard Gaussian law γ_m , independently of its future driving noise. Thus the generator is

$$\mathcal{L}_Q = \text{tr}(Q\nabla^2) - \langle Qx, \nabla \rangle, \quad \text{Cov}(U(\tau), U(\sigma)) = e^{-Q|\tau-\sigma|}.$$

Throughout, τ denotes Ornstein–Uhlenbeck time and t denotes the sequence index.

Lemma 5.1. *Let $D \subset \mathbb{R}^m$ be bounded, nonempty, open and convex. Define*

$$\lambda_Q^{\text{OU}}(D) = \inf_{0 \neq h \in H_0^1(D)} \frac{\int_D \langle Q \nabla h, \nabla h \rangle d\gamma_m}{\int_D h^2 d\gamma_m}.$$

For every $T > 0$,

$$\mathbb{P}\{U([0, T]) \subset D\} \leq e^{-T\lambda_Q^{\text{OU}}(D)}, \quad \lambda_Q(D) \leq \lambda_Q^{\text{OU}}(D) + \frac{1}{2} \text{tr } Q. \quad (43)$$

Proof. The Gaussian density and its reciprocal are bounded on D , so the Gaussian and Lebesgue H_0^1 form domains agree. Integration by parts identifies the displayed form with the killed Ornstein–Uhlenbeck semigroup P_T^D . Its $L^2(\gamma_m)$ operator norm is $e^{-T\lambda_Q^{\text{OU}}(D)}$. The killed-semigroup representation used in Lemma 2.4 therefore gives

$$\mathbb{P}\{U([0, T]) \subset D\} = \langle \mathbf{1}_D, P_T^D \mathbf{1}_D \rangle_{L^2(\gamma_m)} \leq \gamma_m(D) e^{-T\lambda_Q^{\text{OU}}(D)}.$$

This proves the first inequality. Equivalently, the representation follows from the Dirichlet Feynman–Kac formula after the conjugation below; its potential is bounded on D [14].

Write $\rho(x) = (2\pi)^{-m/2} e^{-|x|^2/2}$ and put $g = h\sqrt{\rho}$. For $h \in C_c^\infty(D)$, expansion and integration by parts give

$$\int_D \langle Q \nabla h, \nabla h \rangle d\gamma_m = \int_D \left[\langle Q \nabla g, \nabla g \rangle + \left(\frac{1}{4} x^T Q x - \frac{1}{2} \text{tr } Q \right) g^2 \right] dx.$$

The identity extends by form closure to $H_0^1(D)$. Multiplication by $\sqrt{\rho}$ maps this space bijectively to itself and preserves the corresponding L^2 norms. Since the potential is at least $-\text{tr } Q/2$, taking Rayleigh infima proves the second inequality. $\square$

5.2 One energy interval at one spectral scale

Fix the data of Proposition 2.2 and a positive diagonal Q . We use the matrices and scalar quantities from the preceding section:

$$L_t = R_t Q R_t^T, \quad X_t = L_t / \theta, \quad F_t = F_t^\theta = R_t^T f_0(X_t) R_t, \quad k_t = k_t(\theta) = \text{tr } \phi(X_t).$$

In this subsection the dyadic scale θ is fixed. Recall that $a = 3/2$ and that $E_\theta(s, t)$ is additive and nonnegative for $s \leq t$. Put $h_t = E_\theta(0, t)$.

Lemma 5.2. *There is an absolute c_1 with the following property. Let $\mathcal{I} \subset \{0, \dots, n\}$ be an index set whose values h_t lie in an interval of length at most e_0 , where e_0 is the constant in Lemma 4.1. For every $u \geq 0$,*

$$\mathbb{P} \left\{ \|F_t U(\tau)\| \leq c_1 \sqrt{k_t} \text{ for every } t \in \mathcal{I}, \ u \leq \tau \leq u + \theta^{-1} \right\} \geq \frac{1}{2}. \quad (44)$$

The same conclusion holds on any shorter time interval.

Proof. We may omit indices with $F_t = 0$. By (26), every remaining index has $k_t \geq 1/4$. Divide these indices into the classes

$$\mathcal{I}_j = \{t \in \mathcal{I} : K_j \leq k_t < 2K_j\}, \quad K_j = 2^j/4, \quad j = 0, 1, \dots$$

We first bound the expected supremum within one nonempty class, writing $k = K_j$.

Stationarity, coisometry and (26) give

$$\mathbb{E}\|F_t U(\tau)\|^2 = \text{tr } f_0(X_t)^2 \leq 4k_t \leq 8k. \quad (45)$$

For two indices in the class, Lemma 4.1, with the indices put in chronological order, implies

$$\mathbb{E}\|(F_t - F_s)U(\tau)\|^2 \leq Ck|h_t - h_s|^{1/a}. \quad (46)$$

Here we dropped the positive right weight in that lemma and used $1 + k_t \leq 6k$. For time increments at a fixed index,

$$\text{Cov}(U(\tau) - U(\sigma)) = 2(I - e^{-Q|\tau - \sigma|}) \preceq 2|\tau - \sigma|Q,$$

so that

$$\begin{aligned} \mathbb{E}\|F_t(U(\tau) - U(\sigma))\|^2 &\leq 2|\tau - \sigma| \text{tr}(F_t^2 Q) \\ &= 2\theta|\tau - \sigma| \text{tr}(X_t f_0(X_t)^2) \leq Ck\theta|\tau - \sigma|. \end{aligned} \quad (47)$$

This trace identity uses $R_t R_t^T = I$; it does not require Q and F_t to commute.

We give the elementary chaining argument for the two parameters. Choose H so that $h_t \in [H, H + e_0]$ for every index under consideration, and put $\alpha = (h_t - H)/e_0$ and $\beta = \theta(\tau - u)$ in $[0, 1]$. Split a mixed increment into an index increment and a time increment. Equations (46) and (47) bound its second moment by

$$Ck(|\alpha - \alpha'|^{1/a} + |\beta - \beta'|),$$

where α is the normalized energy coordinate; the constant absorbs $e_0 \leq 1$. At depth l , divide both parameter intervals into dyadic cells of width 2^{-l} and choose a representative from each occupied product cell. There are at most $C4^l$ representatives. The increment from each representative to that of its parent cell is a centered Gaussian vector with second moment at most

$$v_l = Ck2^{-l/a}.$$

For use here, if $G_1, \dots, G_M$ are arbitrarily dependent centered Gaussian vectors and $\mathbb{E}\|G_i\|^2 \leq v$, then

$$\mathbb{E} \max_{i \leq M} \|G_i\| \leq 2\sqrt{v(\log M + 1/2)}. \quad (48)$$

Indeed, for $v > 0$, diagonalizing the covariance of G_i gives $\mathbb{E}e^{\|G_i\|^2/(4v)} \leq e^{1/2}$, because its eigenvalues are nonnegative and sum to at most v . Bound the exponential of the maximum by the sum of the exponentials, then use Jensen's inequality and Cauchy-Schwarz. The case $v = 0$ is immediate.

Apply (48) to the increments at each depth. Starting with (45) and summing the parent increments gives an absolute C_0 such that

$$\mathbb{E} \sup_{\substack{t \in \mathcal{I}_j \\ u \leq \tau \leq u + \theta^{-1}}} \|F_t U(\tau)\| \leq C\sqrt{k} \left(1 + \sum_{l \geq 1} 2^{-l/(2a)} (1 + \sqrt{l}) \right) \leq C_0 \sqrt{k}. \quad (49)$$

To justify convergence of the representatives, note that there are only finitely many distinct energy values. If $h_t = h_s$, Lemma 4.1 gives $F_t = F_s$ exactly. At sufficiently fine depth a target index therefore has the same filter as its representative, and the representative times converge to its target time. Continuity of U completes the passage to the supremum. Equivalently, one may first use finite time grids and then increase them to a dense set. Neither argument uses a lower bound on the positive gaps between energy values. In particular, the number of indices does not enter (49).

It remains to control all the k classes at once. On a finite time grid the supremum S_j in (49) is a maximum of Gaussian vector norms. Represent the joint Gaussian vectors as linear images of one standard Gaussian vector. Each corresponding linear map has norm at most one because its output covariance is $F_t^2 \preceq I$, by (26); consequently S_j is a 1-Lipschitz function of that standard Gaussian vector. Equivalently, it is a supremum of scalar Gaussian functionals, each with variance at most one.

Markov's inequality and (49) give $\mathbb{P}\{S_j \leq 2C_0\sqrt{K_j}\} \geq 1/2$. Borell's Gaussian enlargement inequality [9, Theorem 3.1], applied to this sublevel set, gives

$$\mathbb{P}\{S_j > 2C_0\sqrt{K_j} + r\} \leq 1 - \Phi(r) \leq e^{-r^2/2}, \quad r \geq 0,$$

where Φ is the standard normal distribution function. The same bound holds for the full time supremum by increasing dense grids and path continuity. Choose $c_1 > 2C_0$. The probability that any constraint in class j exceeds $c_1\sqrt{k_t}$ is at most

$$\exp(-(c_1 - 2C_0)^2 K_j / 2).$$

With $A = (c_1 - 2C_0)^2 / 8$, the sum of these bounds is at most

$$\sum_{j \geq 0} e^{-A2^j} \leq \frac{e^{-A}}{1 - e^{-A}} \leq \frac{1}{2}$$

once c_1 is an absolute constant large enough that $A \geq \log 3$. This proves (44). Restricting the time interval can only increase its probability. $\square$

The two bounds in this proof have different roles: the trace of the covariance controls the expected vector norm, while its operator norm gives the absolute concentration parameter. This distinction is what allows the class failure probabilities to be summed without a factor depending on the dimension.

5.3 Combining the constraints

Proof of Proposition 2.2. Fix d, n, Q and the entire recursion. Let Θ be the finite set of dyadic scales for which some F_t^θ is nonzero. If $q_{\min}$ and $q_{\max}$ are the smallest and largest diagonal entries of Q , then coisometry gives $q_{\min}I \preceq L_t \preceq q_{\max}I$, and the support of f_0 gives

$$\Theta \subset [q_{\min}/4, 2q_{\max}], \quad \sum_{\theta \in \Theta} \theta \leq 4q_{\max} \leq 4 \operatorname{tr} Q. \quad (50)$$

At each scale, partition the sequence indices by the half-open intervals

$$[re_0, (r+1)e_0), \quad r = 0, 1, \dots,$$

containing $E_\theta(0, t)$. These are the energy intervals to which Lemma 5.2 applies. There are at most $C(1 + E_\theta(0, n))$ nonempty intervals, including any repetitions of an energy value in the same interval.

For each fixed t , the scale constraints control the original state norm. To see this, diagonalize L_t and write $y = R_t U(\tau)$ in this eigenbasis, with eigenvalues $\lambda_i > 0$. Each λ_i admits a dyadic scale with $\lambda_i/\theta \in [1, 2]$, where $f_0 = 1$; that scale belongs to Θ . Since R_t^T is an isometry,

$$\sum_{\theta \in \Theta} \|F_t^\theta U(\tau)\|^2 = \sum_i |y_i|^2 \sum_{\theta \in \Theta} f_0(\lambda_i/\theta)^2 \geq \|R_t U(\tau)\|^2. \quad (51)$$

Thus all the scale constraints at a given time imply, by (25),

$$\|R_t U(\tau)\|^2 \leq c_1^2 \sum_{\theta \in \Theta} k_t(\theta) \leq c_1^2 C_\phi d \quad (0 \leq t \leq n), \quad (52)$$

with an absolute C_ϕ .

We also impose coordinate bounds. A stationary scalar Ornstein–Uhlenbeck process of rate $q > 0$, restricted to an interval of length at most $1/q$, stays in $[-2, 2]$ with probability at least an absolute $p_0 > 0$. By stationarity and time rescaling, it suffices to consider rate one on $[0, 1]$. In distribution that process is

$$e^{-s}(Z + B(e^{2s} - 1)), \quad 0 \leq s \leq 1,$$

where Z is standard normal and B is an independent standard Brownian motion. The event $|Z| \leq 1$ and $\sup_{r \leq e^2 - 1} |B(r)| \leq 1$ has a fixed positive probability and implies the asserted bound.

Choose absolute constants

$$A_0 > c_1 \sqrt{C_\phi}, \quad B_0 > 2,$$

and let $\mathcal{D}$ be the domain in Proposition 2.2 with these constants. It is bounded, open, centrally symmetric and convex, and contains a neighborhood of zero. The closed constraints (52) and $|U_j(\tau)| \leq 2$ lie strictly inside $\mathcal{D}$.

For a fixed $T > 0$, cover $[0, T]$ by at most $1 + T\theta$ consecutive closed intervals of length at most $1/\theta$ at scale θ . For every such interval and every nonempty energy interval impose the event in Lemma 5.2. Also cover $[0, T]$ by at most $1 + TQ_{jj}$ intervals of length at most $1/Q_{jj}$ for coordinate j and impose its box event. All these events are closed symmetric convex constraints on linear evaluations of the centered Gaussian path. There are finitely many events for this fixed input and T . Lemma 2.5 therefore multiplies their probability lower bounds. In particular, its finite-grid argument and the continuity of the paths apply even when some constraints are duplicates. The strict slack just chosen ensures that their intersection is an event of survival in the open domain $\mathcal{D}$.

It follows that

$$\begin{aligned} -\log \mathbb{P}\{U([0, T]) \subset \mathcal{D}\} &\leq C \sum_{\theta \in \Theta} (1 + T\theta)(1 + E_\theta(0, n)) + C \sum_{j=1}^m (1 + TQ_{jj}) \\ &\leq A(Q, R, d, n) + C'T \operatorname{tr} Q. \end{aligned} \quad (53)$$

Here (28) and (50) bound the coefficient of T by an absolute constant times $\operatorname{tr} Q$, and

$$A(Q, R, d, n) = C \left[\sum_{\theta \in \Theta} (1 + E_\theta(0, n)) + m \right] < \infty.$$

This additive constant may depend on the input and on the smallest entry of Q ; no uniform bound on it is required.

By Lemma 5.1,

$$\lambda_Q^{\text{OU}}(\mathcal{D}) \leq C' \operatorname{tr} Q + \frac{A(Q, R, d, n)}{T}.$$

Let T tend to infinity while keeping the input fixed. This removes the additive constant, including the number of spectral scales. The second inequality in (43) now gives

$$\lambda_Q(\mathcal{D}) \leq (C' + 1/2) \operatorname{tr} Q.$$

Thus $H_0 = C' + 1/2$ is absolute. The constants A_0, B_0, H_0 are independent of d, n, Q and of the recursion, as required. $\square$

6 Further consequences

The signed-prefix theorem gives bounds for other finite-dimensional norms and, through matrix transference, for arrays whose rows can be permuted separately.

6.1 Finite-dimensional ℓ_p bounds

Corollary 6.1 (Finite-dimensional ℓ_p bounds). *Let $1 \leq p \leq \infty$, with $1/\infty = 0$, and let $d, N \geq 1$ be integers. Every sequence $v_1, \dots, v_N \in \mathbb{R}^d$ with $\|v_i\|_p \leq 1$ admits signs $\varepsilon_i \in \{-1, 1\}$ such that*

$$\max_{0 \leq k \leq N} \left\| \sum_{i=1}^k \varepsilon_i v_i \right\|_p \leq C d^{\max\{1/p, 1-1/p\}}.$$

If additionally $\sum_{i=1}^N v_i = 0$, there is a permutation $\pi \in \mathfrak{S}_N$ such that

$$\max_{0 \leq k \leq N} \left\| \sum_{i=1}^k v_{\pi(i)} \right\|_p \leq C d^{\max\{1/p, 1-1/p\}}.$$

Here C is the same absolute constant as in Theorems 1.1 and 1.2, independent of p, d, N . For $1 \leq p \leq 2$, both bounds have sharp dimension order $d^{1/p}$: for every d there is a unit-ball sequence for which every signing has a prefix of norm at least $d^{1/p}$, and a finite indexed zero-sum unit-ball family for which every permutation has a prefix of norm at least $\frac{1}{3}d^{1/p}$.

Proof. For $1 \leq p \leq 2$, the norm comparisons

$$\|x\|_2 \leq \|x\|_p \leq d^{1/p-1/2} \|x\|_2$$

allow direct application of Theorems 1.1 and 1.2, followed by the bound $Cd^{1/p}$ on all prefixes. For $2 \leq p \leq \infty$, use

$$\|x\|_p \leq \|x\|_2 \leq d^{1/2-1/p} \|x\|_p.$$

Divide the input vectors by $d^{1/2-1/p}$, apply the corresponding Euclidean theorem, and scale back to obtain $Cd^{1-1/p}$. Scaling preserves the zero sum, and in each case the one signing or permutation controls all prefixes. At $p = \infty$ this gives Cd .

For the lower bounds, let $1 \leq p \leq 2$. The ordered standard basis $e_1, \dots, e_d$ has final signed sum of norm $d^{1/p}$ for every signing. For the ordering problem, take $e_1, \dots, e_d$ and d copies of $w = -\frac{1}{d}(1, \dots, 1)$. This indexed family sums to zero and lies in the unit ball, since $\|w\|_p = d^{1/p-1} \leq 1$. If $d \geq 2$, consider any permutation and its prefix just after the $k = \lfloor d/2 \rfloor$ -th basis vector. If t copies of w have appeared, put $a = t/d \in [0, 1]$. The prefix sum s has k coordinates equal to $1 - a$ and $d - k$ equal to $-a$. Since both k and $d - k$ are at least $d/3$, convexity gives

$$\|s\|_p^p = k(1-a)^p + (d-k)a^p \geq \frac{d}{3}((1-a)^p + a^p) \geq \frac{d}{3} 2^{1-p}.$$

Thus $\|s\|_p \geq \frac{1}{2}(2d/3)^{1/p} \geq \frac{1}{3}d^{1/p}$, for every permutation. When $d = 1$, the family is $(1, -1)$ and either permutation has a prefix of norm one. $\square$

6.2 A colorful row-permutation consequence

Bárány's matrix transference principle turns the signed-prefix bound into a row-permutation bound for arrays. Each row may have its own permutation, while the controlled sums aggregate the same number of entries from every row.

Corollary 6.2 (Euclidean colorful Steinitz bound). *Let $d, k, n \geq 1$ be integers and let $a_i^j \in \mathbb{R}^d$, for $1 \leq j \leq k$ and $1 \leq i \leq n$, satisfy*

$$\|a_i^j\|_2 \leq 1, \quad \sum_{j=1}^k \sum_{i=1}^n a_i^j = 0.$$

There is one tuple of permutations $(\pi_1, \dots, \pi_k) \in \mathfrak{S}_n^k$ such that

$$\max_{0 \leq m \leq n} \left\| \sum_{j=1}^k \sum_{i=1}^m a_{\pi_j(i)}^j \right\|_2 \leq 2C\sqrt{d},$$

where C is the absolute constant in Theorem 1.1. The same tuple works for every m and may be chosen jointly from the whole array. Only the global total sum is assumed to vanish; the bound controls the displayed aggregate synchronous prefixes, not each row separately.

Proof. For an arbitrary $k \times n$ array $B = (b_i^j)$ in the Euclidean unit ball B_2^d , with no zero-sum condition, list its entries column by column:

$$v_{(i-1)k+j} = b_i^j \quad (1 \leq i \leq n, 1 \leq j \leq k).$$

Apply Theorem 1.1 to this sequence of length kn . With $\varepsilon_i^j = \varepsilon_{(i-1)k+j}$, every complete-column prefix satisfies

$$\left\| \sum_{j=1}^k \sum_{i=1}^m \varepsilon_i^j b_i^j \right\|_2 = \left\| \sum_{\ell=1}^{km} \varepsilon_\ell v_\ell \right\|_2 \leq C\sqrt{d}.$$

This is the column-major reduction in the proof of [7, Lemma 2.1], with the present signed-prefix bound in place of its dimension-linear bound. Thus Bárány's matrix signed constant satisfies $V(B_2^d) \leq C\sqrt{d}$, uniformly in k, n . Here V allows an entrywise signing of an arbitrary array, whereas the row-permutation constant U ranges over total-zero arrays and controls the same aggregate prefixes. His transference theorem [7, Theorem 2.2] gives $U(B_2^d) \leq 2V(B_2^d)$ for every number of rows. Shorter auxiliary arrays may be padded with trailing zero columns when using the bound uniformly in the number of columns. Since B_2^d is a centrally symmetric convex body, this gives the asserted bound. $\square$

Taking $k = 1$ in the regular-simplex example in the introduction shows that no $o(\sqrt{d})$ bound can hold uniformly over these arrays. Thus the order in the dimension is sharp; the numerical constant $2C$ is not asserted to be optimal. The row permutations are existential, with no online or efficient algorithm claimed.

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